

# MUHAMMAD AHMED MUFTI

Software Engineer

[m.ahmed.mufti@gmail.com](mailto:m.ahmed.mufti@gmail.com) | +92 321 8547417 | Islamabad, Pakistan  
[ahmedmufti.github.io/portfolio](https://ahmedmufti.github.io/portfolio) | [github.com/AhmedMufti](https://github.com/AhmedMufti) | [linkedin.com/in/m-ahmed-m-26491913a](https://linkedin.com/in/m-ahmed-m-26491913a)

## SUMMARY

Software engineer across the stack: six production web apps live today (React, Next.js, Node.js, FastAPI, PostgreSQL, MongoDB), an event-driven Java and Kafka service, and a sole-author research study of small language models in agentic RAG with open-source code. Computer Science graduate, FAST-NUCES (CGPA 3.6 / 4.00, Dean's List four semesters), Fiverr Level 1 Seller, former Computer Networks teaching assistant.

## TECHNICAL SKILLS

|                   |                                                                                                     |
|-------------------|-----------------------------------------------------------------------------------------------------|
| Languages         | JavaScript / TypeScript, Python, Java, SQL, C/C++, Kotlin                                           |
| Web               | React, Next.js, Redux Toolkit, Tailwind CSS, Node.js, Express, FastAPI, REST, JWT / RBAC, Socket.io |
| Backend & Data    | Spring Boot, Apache Kafka, PostgreSQL, MongoDB, MySQL, SQL Server, DuckDB, Mongoose                 |
| AI / ML           | PyTorch, Hugging Face Transformers, LangChain, LangGraph, RAG, FAISS, Modal (serverless GPU)        |
| Practices & Tools | Git, Docker Compose, Testcontainers, JUnit 5, Postman, Vercel, Linux, OWASP Top 10, Lighthouse      |

## SELECTED PROJECTS

### reccs.co: Product Recommendation Platform **LIVE** 2025

React, FastAPI, DuckDB, Modal GPU, fine-tuned RoBERTa. Live: [reccs.co](https://reccs.co)

- Ranks products from sentiment across 14.1 million Reddit comments in 937 subreddits: fine-tuned RoBERTa classification on Modal GPU workers, a FastAPI and DuckDB API, and a React front end with search, grades, and guides.

### MedScholar: Clinical Research Platform (Final Year Project) **LIVE** 2025 - 2026

React, FastAPI, PostgreSQL, LangChain, FAISS. Live: [med-scholar-ten.vercel.app](https://med-scholar-ten.vercel.app)

- Retrieval-augmented workspace that answers clinical questions from PubMed literature with cited sources, using a FAISS index with strict grounding constraints to limit hallucination.

### IndonesiaCompliance: Regulatory Roadmap Platform **LIVE** 2024 - 2025

Next.js, Node.js, PostgreSQL, Vercel. Live: [indonesia-compliance.vercel.app](https://indonesia-compliance.vercel.app)

- Production platform that has generated over 1,000 compliance roadmaps from Indonesian government data, with KBLI code search and license guides on a Next.js front end over a PostgreSQL-backed API.

### Small Language Models in Agentic RAG (Independent Research, Sole Author) 2026

Python, Modal serverless GPUs, BM25 + BGE retrieval, ReAct, LLM-as-judge. Code (MIT): [github.com/AhmedMufti/slm-agentic-financial-rag](https://github.com/AhmedMufti/slm-agentic-financial-rag)

- Benchmarked four small language models (1.7B to 8B) and a 117B ceiling on FinanceBench under naive RAG and an agentic ReAct loop across 1,499 cells; built the full harness, a 10-category failure taxonomy, and a two-judge pipeline (Cohen's kappa 0.65) for under USD 25 of compute.

### Order Pipeline: Two Spring Boot Services over Kafka 2026

Java, Spring Boot, Spring Kafka, Docker Compose, Testcontainers.

- order-service publishes OrderCreated events to a three-partition topic; inventory-service consumes idempotently and dead-letters bad events. Verified rebalancing across instances, replay without reprocessing, and fan-out to a second consumer group, with Testcontainers integration tests.

### Pakistan Legal Nexus: Legal Operations Platform **LIVE** 2025 - 2026

MERN: React, Redux Toolkit, i18next, Express, MongoDB, JWT, Socket.io. Live: [earnest-praline-2c72a6.netlify.app](https://earnest-praline-2c72a6.netlify.app)

- Role-based platform for lawyers, litigants, and clerks with case files, document upload, lawyer directory, marketplace, and real-time chat in English and Urdu; 83 API endpoints over 16 models.

## EXPERIENCE

### Freelance Software Engineer, Fiverr (Level 1 Seller) Jun 2024 - Present

Independent clients, remote

- Delivered full-stack web and machine-learning projects for international clients, taking each from brief to deployed product, and reached Level 1 Seller through consistent on-time delivery.
- Built Brand Engine, a generative-AI service that produces brand assets on demand (Python, FastAPI, ComfyUI with Flux.1 Schnell, Modal serverless GPUs).

### Teaching Assistant (Part-time) Jan 2025 - Jun 2026

FAST-NUCES, Islamabad, Pakistan

- Ran labs, wrote assignments, and graded projects for Computer Networks, Theory of Automata, and Professional Practices in IT, supporting over 100 students; selected on academic performance.

### Cyber Security Intern Jul 2025 - Aug 2025

Risk Associates, Islamabad, Pakistan

- Ran penetration tests and security audits on enterprise web applications and reported OWASP Top 10 vulnerabilities with remediation guidance.

## EDUCATION

### BSc Computer Science Aug 2022 - Jun 2026

National University of Computer and Emerging Sciences (FAST-NUCES), Islamabad, Pakistan

- CGPA 3.6 / 4.00. Dean's List for four consecutive semesters (Spring 2024 to Fall 2025). Coursework: Software Engineering (A+), Operating Systems (A+), Computer Networks (A), Database Systems, Software Design and Analysis, Compiler Construction, Generative AI (A+), Agentic AI (A+).

## ACHIEVEMENTS & CERTIFICATIONS

- Dean's List, four consecutive semesters. Fiverr Level 1 Seller. IELTS Academic 8.0. Machine Learning Specialization (Stanford / Coursera, in progress); Pre Security (TryHackMe).